Project #2

World Happiness Report K-Means Clustering

Minh Truong & Mallie Lange

Dataset Information + K-Means Clustering

The dataset was chosen from kaggle.com.

The World Happiness Report is a survey of global happiness. The six factors are: economic production, social support, life expectancy, freedom, absence of corruption, and generosity. Each row represents the average scores for that specific country. The overall happiness score (the ladder score) is the sum of six factors and the 'dystopia + residual'.

We chose to utilize the unsupervised learning method of K-Means clustering to group the countries based on similarities across the multiple factors.

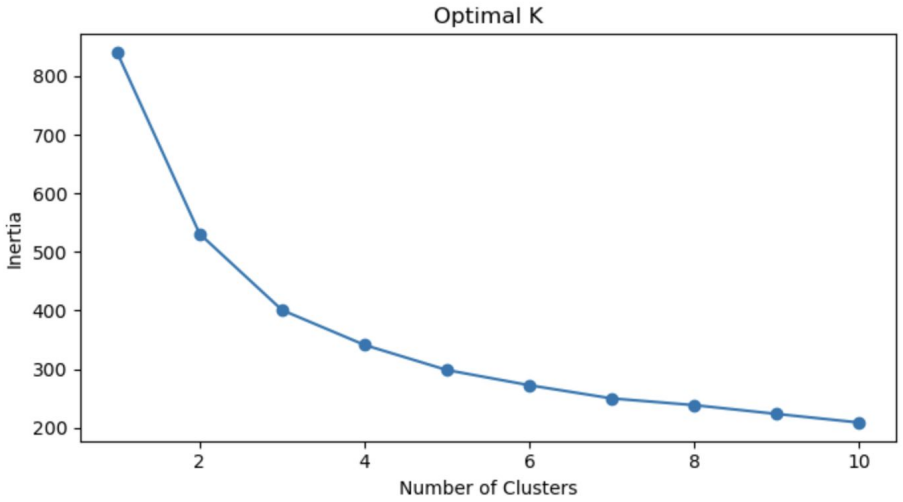
```
RangeIndex: 143 entries, 0 to 142
Data columns (total 12 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   Country name                  143 non-null    object
 1   Regional indicator            143 non-null    object
 2   Ladder score                  143 non-null    float64
 3   upperwhisker                  143 non-null    float64
 4   lowerwhisker                  143 non-null    float64
 5   Log GDP per capita            140 non-null    float64
 6   Social support                140 non-null    float64
 7   Healthy life expectancy       140 non-null    float64
 8   Freedom to make life choices  140 non-null    float64
 9   Generosity                    140 non-null    float64
 10  Perceptions of corruption     140 non-null    float64
 11  Dystopia + residual           140 non-null    float64
dtypes: float64(10), object(2)
memory usage: 13.5+ KB
None
       Ladder score  ...  Dystopia + residual
count    143.000000  ...           140.000000
mean       5.527580  ...             1.575914
std        1.170717  ...             0.537459
min        1.721000  ...            -0.073000
25%        4.726000  ...             1.308250
50%        5.785000  ...             1.644500
75%        6.416000  ...             1.881750
max        7.741000  ...             2.998000
```

K-Means Clustering - Elbow Method

Using the Elbow method, we chose three clusters of best grouped countries based on the factors.

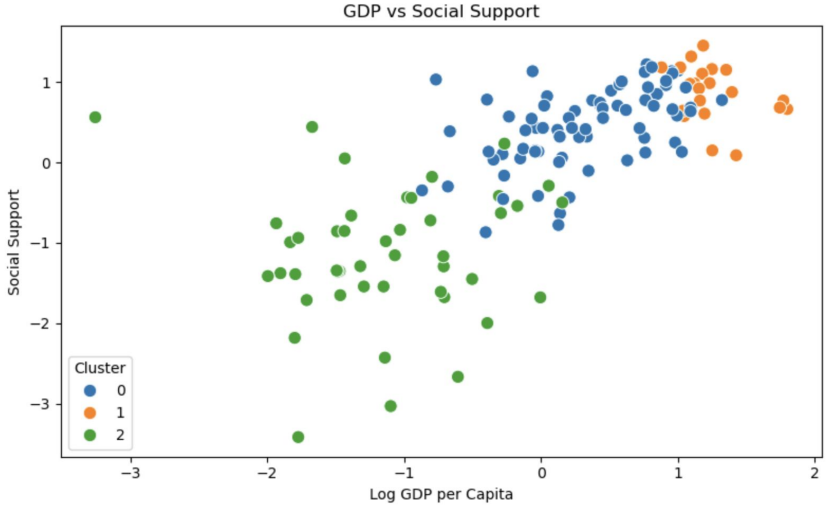
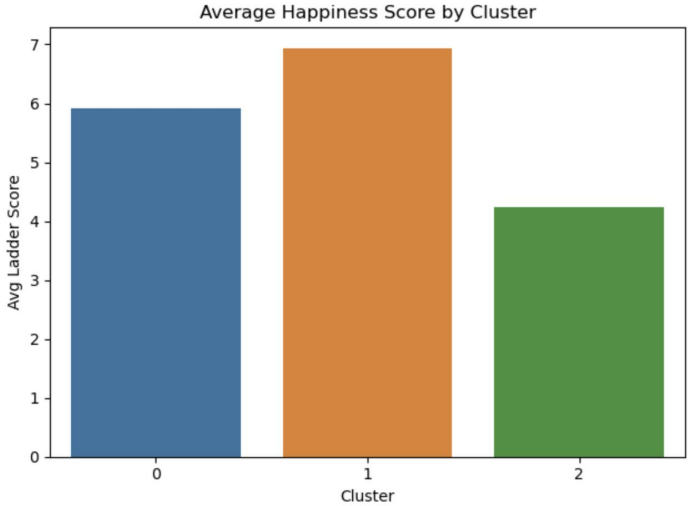
The three clusters are:

- High levels of happiness: Countries that are doing really well in terms of strong economies, good healthcare, lots of social support and people feel more free
- Moderate levels of happiness: Countries that are going okay with a moderate quality of life and mixed results for the features
- Low levels of happiness: Countries that are not doing very well with lower incomes, weaker healthcare, and less social support



Visualization + Conclusion

The scatter plot shows countries grouped into three clusters based on their log GDP per capita and social support, with each color representing one of the clusters. This scatterplot is one example of how the countries naturally are clustered.



The bar chart validates the cluster labels by comparing average ladder scores, where Cluster 1 ranks highest (~6.95), Cluster 0 falls in the middle (~5.95), and Cluster 2 ranks lowest (~4.25). Using K-Means clustering with k=3, 143 countries were grouped into three happiness segments: high, medium, and low, with GDP per capita and social support identified as the primary differentiating factors.